

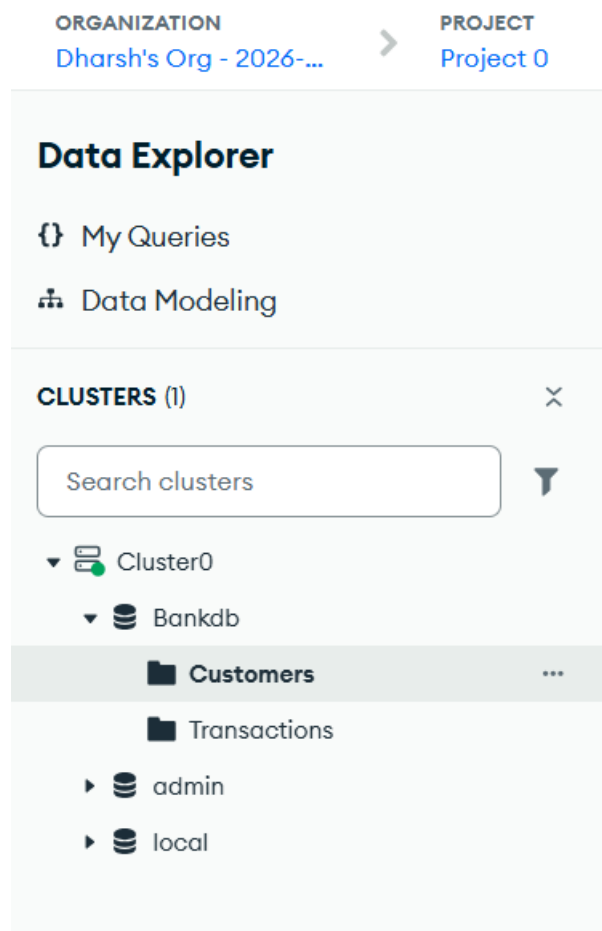
DATABASE SYSTEM ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT

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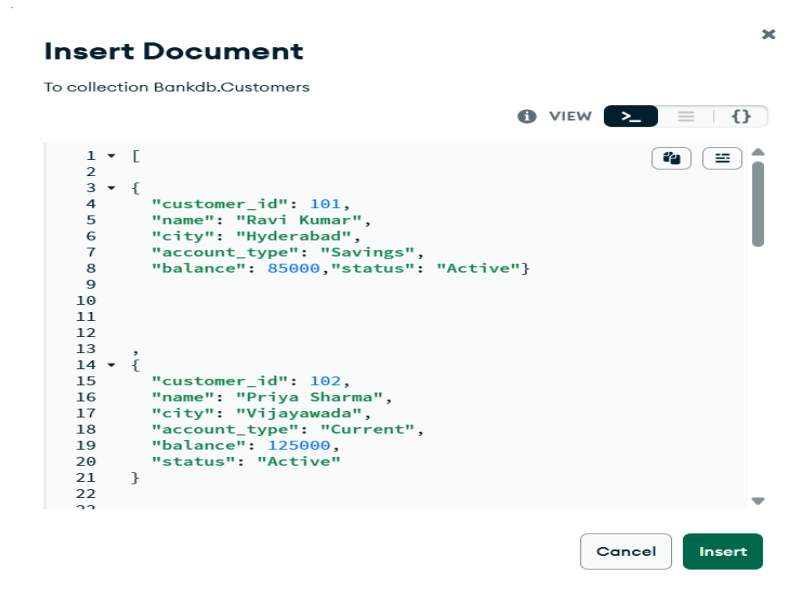
Step 1: Create the Database and First Collection

Database Name : bankdb
Collection Name : customers



Step 2: Insert Customer Data (8 Documents)

Open bankdb → customers → Documents. Click Add Data → Insert Document. Insert each of the following JSON documents one at a time, then click Insert.



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Data Explorer

My Queries, Data Modeling

CLUSTERS (1)

Cluster0

Bankdb

Customers

Transactions

admin

local

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Cluster0 > Bankdb > Customers

Documents 8 | Aggregations | Schema | Indexes 1 | Validation

Type a query: { field: 'value' } or [Generate query](#)

EXPLAIN | RESET | FIND | OPTIONS

ADD DATA | BULK | EXPORT CODE

25 | 1 - 8 of 8

	_id ObjectId	customer_id Int32	name String	city String	account_type String	
1	ObjectId('6aad99660ce5083...')	101	"Ravi Kumar"	"Hyderabad"	"Savings"	
2	ObjectId('6aad99660ce5083...')	102	"Priya Sharma"	"Vijayawada"	"Current"	
3	ObjectId('6aad99660ce5083...')	103	"Arjun Reddy"	"Hyderabad"	"Savings"	
4	ObjectId('6aad99660ce5083...')	104	"Sneha Rao"	"Warangal"	"Savings"	
5	ObjectId('6aad99660ce5083...')	105	"Kiran Kumar"	"Guntur"	"Current"	
6	ObjectId('6aad99660ce5083...')	106	"Anjali Singh"	"Hyderabad"	"Savings"	
7	ObjectId('6aad99660ce5083...')	107	"Rahul Verma"	"Bengaluru"	"Current"	
8	ObjectId('6aad99660ce5083...')	108	"Meena Devi"	"Chennai"	"Savings"	

Step 3 : Insert Transaction Data (15 Documents)

Open bankdb → transactions → Documents. Click Add Data → Insert Document. Insert each transaction JSON one at a time.

```

{ "transaction_id": 1001, "customer_id": 101, "type": "Deposit",
  "amount": 25000, "mode": "UPI", "date": "2026-08-01", "status": "Success" }

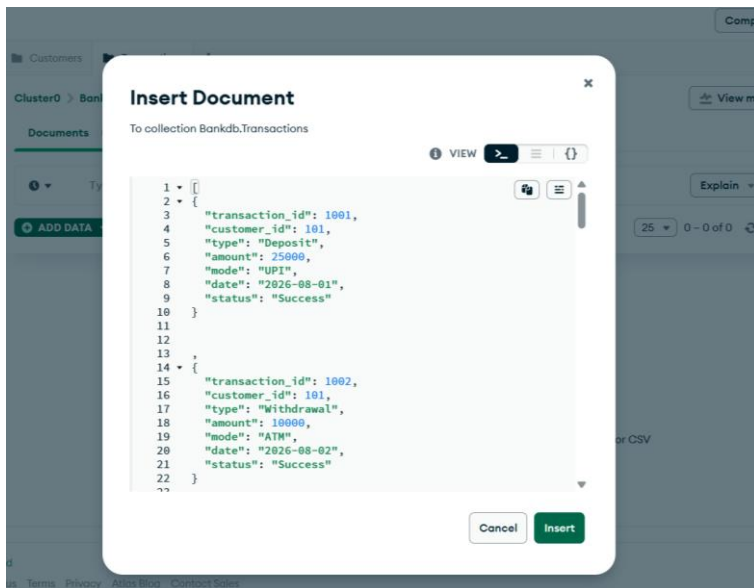
{ "transaction_id": 1002, "customer_id": 101, "type": "Withdrawal",
  "amount": 10000, "mode": "ATM", "date": "2026-08-02", "status": "Success" }

{ "transaction_id": 1003, "customer_id": 102, "type": "Deposit",
  "amount": 50000, "mode": "NEFT", "date": "2026-08-02", "status": "Success" }

{ "transaction_id": 1004, "customer_id": 102, "type": "Withdrawal",
  "amount": 15000, "mode": "ATM", "date": "2026-08-03", "status": "Success" }

```

```
{ "transaction_id": 1005, "customer_id": 103, "type": "Deposit",  
  "amount": 30000, "mode": "UPI", "date": "2026-08-03", "status": "Success" }  
  
{ "transaction_id": 1006, "customer_id": 103, "type": "Withdrawal",  
  "amount": 5000, "mode": "ATM", "date": "2026-08-04", "status": "Success" }  
  
{ "transaction_id": 1007, "customer_id": 104, "type": "Deposit",  
  "amount": 100000, "mode": "NEFT", "date": "2026-08-04", "status": "Success" }  
  
{ "transaction_id": 1008, "customer_id": 104, "type": "Withdrawal",  
  "amount": 25000, "mode": "ATM", "date": "2026-08-05", "status": "Success" }  
  
{ "transaction_id": 1009, "customer_id": 105, "type": "Deposit",  
  "amount": 15000, "mode": "UPI", "date": "2026-08-05", "status": "Success" }  
  
{ "transaction_id": 1010, "customer_id": 106, "type": "Deposit",  
  "amount": 75000, "mode": "NEFT", "date": "2026-08-06", "status": "Success" }  
  
{ "transaction_id": 1011, "customer_id": 106, "type": "Withdrawal",  
  "amount": 20000, "mode": "ATM", "date": "2026-08-06", "status": "Success" }  
  
{ "transaction_id": 1012, "customer_id": 107, "type": "Deposit",  
  "amount": 40000, "mode": "IMPS", "date": "2026-08-07", "status": "Success" }  
  
{ "transaction_id": 1013, "customer_id": 107, "type": "Withdrawal",  
  "amount": 12000, "mode": "ATM", "date": "2026-08-07", "status": "Success" }  
  
{ "transaction_id": 1014, "customer_id": 108, "type": "Deposit",  
  "amount": 150000, "mode": "NEFT", "date": "2026-08-08", "status": "Success" }  
  
{ "transaction_id": 1015, "customer_id": 108, "type": "Withdrawal",  
  "amount": 30000, "mode": "ATM", "date": "2026-08-08", "status": "Success" }
```



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Documents 15 | Aggregations | Schema | Indexes 1 | Validation

Type a query: { field: 'value' } or [Generate query](#)

EXPLAIN | RESET

ADD DATA | BULK | EXPORT CODE

25 | 1 - 15 of 15

	_id ObjectId	transaction_id Int32	customer_id Int32	type String	amount Int32
1	ObjectId('6aad0b240ce5083...')	1001	101	"Deposit"	25000
2	ObjectId('6aad0b240ce5083...')	1002	101	"Withdrawal"	10000
3	ObjectId('6aad0b240ce5083...')	1003	102	"Deposit"	50000
4	ObjectId('6aad0b240ce5083...')	1004	102	"Withdrawal"	15000
5	ObjectId('6aad0b240ce5083...')	1005	103	"Deposit"	30000
6	ObjectId('6aad0b240ce5083...')	1006	103	"Withdrawal"	5000
7	ObjectId('6aad0b240ce5083...')	1007	104	"Deposit"	100000
8	ObjectId('6aad0b240ce5083...')	1008	104	"Withdrawal"	25000
9	ObjectId('6aad0b240ce5083...')	1009	105	"Deposit"	15000

Basic Find Queries (Q1 – Q11)

For each query below: open the collection → Documents tab → paste the filter JSON into the Filter box → click Find.

Query 1: Display All Customers

Collection: customers

```
{}
```

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ADD DATA BULK EXPORT CODE

25 1 - 8 of 8

	_id ObjectId	customer_id Int32	name String	city String	account_type String
1	ObjectId('6aad99660ce5083_101')	101	"Ravi Kumar"	"Hyderabad"	"Savings"
2	ObjectId('6aad99660ce5083_102')	102	"Priya Sharma"	"Vijayawada"	"Current"
3	ObjectId('6aad99660ce5083_103')	103	"Arjun Reddy"	"Hyderabad"	"Savings"
4	ObjectId('6aad99660ce5083_104')	104	"Sneha Rao"	"Warangal"	"Savings"
5	ObjectId('6aad99660ce5083_105')	105	"Kiran Kumar"	"Guntur"	"Current"
6	ObjectId('6aad99660ce5083_106')	106	"Anjali Singh"	"Hyderabad"	"Savings"
7	ObjectId('6aad99660ce5083_107')	107	"Rahul Verma"	"Bengaluru"	"Current"
8	ObjectId('6aad99660ce5083_108')	108	"Meena Devi"	"Chennai"	"Savings"

Query 2: Find Active Customers

Filter:

```
{
  "status": "Active"
}
```

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ADD DATA BULK EXPORT CODE

25 1 - 7 of 7

	_id ObjectId	customer_id Int32	name String	city String	account_type String
1	ObjectId('6aad99660ce5083_101')	101	"Ravi Kumar"	"Hyderabad"	"Savings"
2	ObjectId('6aad99660ce5083_102')	102	"Priya Sharma"	"Vijayawada"	"Current"
3	ObjectId('6aad99660ce5083_103')	103	"Arjun Reddy"	"Hyderabad"	"Savings"
4	ObjectId('6aad99660ce5083_104')	104	"Sneha Rao"	"Warangal"	"Savings"
5	ObjectId('6aad99660ce5083_106')	106	"Anjali Singh"	"Hyderabad"	"Savings"
6	ObjectId('6aad99660ce5083_107')	107	"Rahul Verma"	"Bengaluru"	"Current"
7	ObjectId('6aad99660ce5083_108')	108	"Meena Devi"	"Chennai"	"Savings"

Query 3: Find Customers With Balance Greater Than 100000

Filter:

```
{
  "balance": {
    "$gt": 100000
  }
}
```

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Project 0

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Cluster0 > Bankdb > Customers

Documents 8 Aggregations Schema Indexes 1 Validation

`{ "balance": { "$gt": 100000 } }`

Generate query + Explain Reset Fit

ADD DATA BULK EXPORT CODE

25 1 - 4 of 4

	_id ObjectId	customer_id Int32	name String	city String	account_type String
1	ObjectId('6aad09660ce5083...')	102	Priya Sharma	Vijayawada	Current
2	ObjectId('6aad09660ce5083...')	104	Sneha Rao	Warangal	Savings
3	ObjectId('6aad09660ce5083...')	106	Anjali Singh	Hyderabad	Savings
4	ObjectId('6aad09660ce5083...')	108	Meena Devi	Chennai	Savings

Query 4: Find Hyderabad Customers

Filter:

```
{
  "city": "Hyderabad"
}
```

Customers Transactions +

Cluster0 > Bankdb > Customers

Documents 8 Aggregations Schema Indexes 1 Validation

`{ "city": "Hyderabad" }`

Generate query + Explain Reset Fit

ADD DATA BULK EXPORT CODE

25 1 - 3 of 3

	_id ObjectId	customer_id Int32	name String	city String	account_type String
1	ObjectId('6a8d679c8c540e0...')	101	Ravi Kumar	Hyderabad	Savings
2	ObjectId('6a8d679c8c540e0...')	103	Arjun Reddy	Hyderabad	Savings
3	ObjectId('6a8d679c8c540e0...')	106	Anjali Singh	Hyderabad	Savings

Query 5: Find Savings Account Customers

Filter:

```
{
  "account_type": "Savings"
}
```

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- Data Modeling

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 - Transactions
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Cluster0 > Bankdb > Customers

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Documents 8 Aggregations Schema Indexes 1 Validation

{

"account_type": "Savings"

}

Generate query + Explain Reset Fin

ADD DATA BULK EXPORT CODE

25 1 - 5 of 5

Customers

	_id ObjectId	customer_id Int32	name String	city String	account_type String
1	ObjectId('6a8d679c8c540e0...	101	"Ravi Kumar"	"Hyderabad"	"Savings"
2	ObjectId('6a8d679c8c540e0...	103	"Arjun Reddy"	"Hyderabad"	"Savings"
3	ObjectId('6a8d679c8c540e0...	104	"Sneha Rao"	"Warangal"	"Savings"
4	ObjectId('6a8d679c8c540e0...	106	"Anjali Singh"	"Hyderabad"	"Savings"
5	ObjectId('6a8d679c8c540e0...	108	"Meena Devi"	"Chennai"	"Savings"

Query 6: Find Transactions Greater Than 20000

Open:

```
transactions
|
+---- Documents
```

Filter:

```
{
  "amount": {
    "$gt": 20000
  }
}
```

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- Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - Bankdb
 - Customers
 - Transactions
 - admin
 - local
 - sample_mflix

Customers Transactions +

Cluster0 > Bankdb > Transactions

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Documents 15 Aggregations Schema Indexes 1 Validation

{

"amount": {

"\$gt": 20000

}

}

Generate query + Explain Reset Fin

ADD DATA BULK EXPORT CODE

25 1 - 9 of 9

Transactions

	String	amount Int32	mode String	date String	status String
1	sit"	25000	"UPI"	"2026-08-01"	"Success"
2	sit"	50000	"NEFT"	"2026-08-02"	"Success"
3	sit"	30000	"UPI"	"2026-08-03"	"Success"
4	sit"	100000	"NEFT"	"2026-08-04"	"Success"
5	drawal"	25000	"ATM"	"2026-08-05"	"Success"
6	sit"	75000	"NEFT"	"2026-08-06"	"Success"

Query 7: Find Deposits

Filter:

```
{
  "type": "Deposit"
}
```

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CLUSTERS (1)

Search clusters

Cluster0

- Bankdb
 - Customers
 - Transactions
- admin
- local
- sample_mflix

Cluster0 > Bankdb > Transactions

Documents 15 Aggregations Schema Indexes 1 Validation

Generate query + Explain Res

ADD DATA BULK EXPORT CODE

25 1 - 8 of 8

_id	ObjectId	transaction_id	customer_id	type	amount
1	ObjectId('6a8d695d8c540e0...')	1001	101	"Deposit"	25000
2	ObjectId('6a8d695d8c540e0...')	1003	102	"Deposit"	50000
3	ObjectId('6a8d695d8c540e0...')	1005	103	"Deposit"	30000
4	ObjectId('6a8d695d8c540e0...')	1007	104	"Deposit"	100000
5	ObjectId('6a8d695d8c540e0...')	1009	105	"Deposit"	15000
6	ObjectId('6a8d695d8c540e0...')	1010	106	"Deposit"	75000
7	ObjectId('6a8d695d8c540e0...')	1012	107	"Deposit"	40000

Query 8: Find Atm Transactions

Filter:

```
{
  "mode": "ATM"
}
```

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CLUSTERS (1)

Search clusters

Cluster0

- Bankdb
 - Customers
 - Transactions
- admin
- local
- sample_mflix

Cluster0 > Bankdb > Transactions

Documents 15 Aggregations Schema Indexes 1 Validation

Generate query + Explain Reset

ADD DATA BULK EXPORT CODE

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_id	ObjectId	transaction_id	customer_id	type	amount
1	ObjectId('6a8d695d8c540e0...')	1002	101	"Withdrawal"	10000
2	ObjectId('6a8d695d8c540e0...')	1004	102	"Withdrawal"	15000
3	ObjectId('6a8d695d8c540e0...')	1006	103	"Withdrawal"	5000
4	ObjectId('6a8d695d8c540e0...')	1008	104	"Withdrawal"	25000
5	ObjectId('6a8d695d8c540e0...')	1011	106	"Withdrawal"	20000
6	ObjectId('6a8d695d8c540e0...')	1013	107	"Withdrawal"	12000
7	ObjectId('6a8d695d8c540e0...')	1015	108	"Withdrawal"	30000

Query 9: Find Deposits Greater Than 50000

Filter:

```
{
  "type": "Deposit",
  "amount": {
```

```

"$gt": 50000
}

```

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My Queries
Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - Bankdb
 - Customers
 - Transactions

Cluster0 > Bankdb > Transactions

Documents 15 Aggregations Schema Indexes 1 Validation

```

{
  "type": "Deposit",
  "amount": {
    "$gt": 50000
  }
}

```

Generate query Explain Reset Find

ADD DATA BULK EXPORT CODE

25 1-3 of 3

	id String	amount Int32	mode String	date String	status String
1	posit	100000	"NEFT"	"2026-08-04"	"Success"
2	posit	75000	"NEFT"	"2026-08-06"	"Success"
3	posit	150000	"NEFT"	"2026-08-08"	"Success"

Query 10: Find Hyderabad Customers With High Balance

Filter:

```

{
  "city": "Hyderabad",
  "balance": {
    "$gt": 100000
  }
}

```

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - Bankdb
 - Customers
 - Transactions

Cluster0 > Bankdb > Customers

Documents 8 Aggregations Schema Indexes 1 Validation

```

{
  "city": "Hyderabad",
  "balance": {
    "$gt": 100000
  }
}

```

Generate query Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1-1 of 1

	_id ObjectId	customer_id Int32	name String	city String	account_type String
1	ObjectId('6a8d679c8c549e8...')	106	"Anjali Singh"	"Hyderabad"	"Savings"

Query 11: Find Customers In Multiple Cities

Filter:

```

{
  "city": {
    "$in": [
      "Hyderabad",
      "Vijayawada",
      "Warangal"
    ]
  }
}

```

❧ Data Modeling

Search clusters

▼  Cluster0

▼ Bankdb

- Customers
- Transactions

- ▶  admin

▶ local

- ▶  sample_mflix

Customers

- Transactions

+

Cluster0 > Bankdb > Customers

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Documents 8

Aggregations

Scheme 1

Indexes 1

Validation


```
{
  "city": {
    "$in": [
      "Hyderabad",
      "Vijayawada",
      "Warangal"
    ]
  }
}
```

[Generate query](#) ⚡

Explain ▼

Resc

ADD DATA

 **BULK** ▼

[EXPORT CODE](#)

25

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🏠 Customers

	Id	customer_id Int32	name String	city String	account_type String
1	6a8d679c8c540ee0	101	"Ravi Kumar"	"Hyderabad"	"Savings"
2	6a8d679c8c540ee0	102	"Priya Sharma"	"Vijayawada"	"Current"
3	6a8d679c8c540ee0	103	"Arjun Reddy"	"Hyderabad"	"Savings"
4	6a8d679c8c540ee0	104	"Sneha Rao"	"Warangal"	"Savings"